O SERIE DI POTENZE

$$\sum_{n=0}^{\infty} a_n (x-x_0)^n$$

$(a_n) \in \mathbb{R}$



DEF. RAGGIO DI CONVERGENZA

$r = \{$ SUPERIORE x: $\sum a_n (x-x_0)^n$ CONVERGE $\}$

$r = 0$ $(x = x_0)$

$r = \infty$ (CONVERGE DAPERTUTTO)

$0 < r < \infty$ $]x_0 - r, \; x_0 + r[$



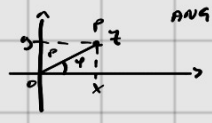
$(\mathbb{C}, +, \cdot)$

$\sum a_n (z-z_0)^n$

NUMERI COMPLESSI

PIANO DI GAUSS

$z = x + iy$, $(x,y) \in \mathbb{R}$, $i^2 = -1$



TEOREMA $\sum a_n x^n$ $(x_0 = 0)$ $x_0 \in \mathbb{R}$

SE LA SERIE CONVERGE IN x_1, ALLORA CONVERGERÀ PER TUTTI GLI $|x| < |x_1|$

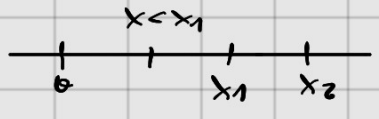
DIMOSTRAZIONE $\sum a_n x_1^n$ CONVERGE $\Rightarrow$ (CONDIZIONE NECESSARIA CRITERIO DI CAUCHY) $\lim_{n\to\infty} a_n x_1^n = 0$

$$|a_n x^n| = |a_n x_1^n| \cdot \left|\frac{x^n}{x_1^n}\right| < \varepsilon \quad q^n \quad q = \frac{x}{x_1} < 1$$

↓ CRITERIO DI WEIERSTRASS

SERIE ARMONICA ??

$\begin{cases} \text{SE } \lim_{n\to\infty} a_n = 0 \\ \not\Rightarrow \text{ CONVERGENTE} \\ \sum \frac{1}{n}, \sum \frac{1}{n^2} \end{cases}$



SE DIVERGE IN x_2

DIVERGE $\forall |x| > |x_2|$

$\sum a_n x_2^n$ DIVERGE

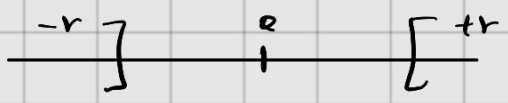
CALCOLO DELL'INTERVALLO DI CONVERGENZA r

(
↓

TEOREMA/CRITERIO DEL RAPPORTO

$\sum a_n x^n$ SE $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \ell \Rightarrow r = \frac{1}{\ell}$

a_n: COEFF. DELLA SERIE A POTENZE



DIMOSTRAZIONE:

$$\left|\frac{a_{n+1}\,x^{n+1}}{a_n\,x^n}\right| = |x| \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = |x|\cdot\ell$$

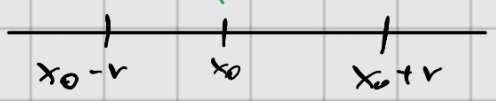
ASSOLUTA CONVERGENZA

$|x|\cdot\ell < 1$

$|x| < \frac{1}{\ell} = r$



$\sum a_n (x-x_0)^n$



$|x-x_0| < r$

$\sum a_n w^n \quad w = (x-x_0) \quad |w| < r \quad -r < w < r$

PER SOSTITUZIONE

$\ell = 0 \quad \frac{1}{\ell} = \infty$ (CONVERGE CON $\forall x$)

$\ell = \infty \quad \frac{1}{\ell} = 0$ (SERIE NON CONVERGE (ECCETTO IN $x=0$))

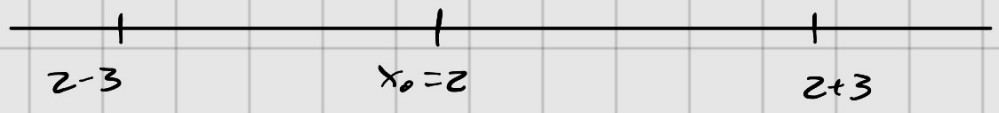
ES. 1

$\sum \frac{(x-2)^n}{3^n} \qquad a_n = \frac{1}{3^n} \qquad x_0 = 2$

$\sum \frac{1}{3^n} w^n \qquad w = x-2 \qquad \lim_{n\to\infty}\left|\frac{\frac{1}{3^{n+1}}}{\frac{1}{3^n}}\right| = \lim_{n\to\infty}\frac{3^n}{3^{n+1}} = \frac{1}{3} = \ell$

$\ell = \frac{1}{3}$ $r = 3$ è l'inverso di ℓ $\qquad \ell < 1 \quad |w| < 3 \quad r = 3$

$-3 < w < 3 \quad -3 < x-2 < 3$



$$I\ ]-1;5[$$

SE L'INTERVALLO DI CONVERGENZA È CHIUSO?

$x=5 \longrightarrow \sum \frac{(5-2)^n}{3^n} = \sum 1$ DIVERGE

$x=-1$

$$\sum \frac{(-1-2)^n}{3^n} \longrightarrow \sum \frac{(-1)^n 3^n}{3^n} = \text{INDETERMINATO}$$

ES. 2

$$\sum \frac{(n!)^2}{(2n)!}(x-1)^n \qquad a_n = \frac{(n!)^3}{(2n)!}$$

$(n+1)! = (n+1)n!$

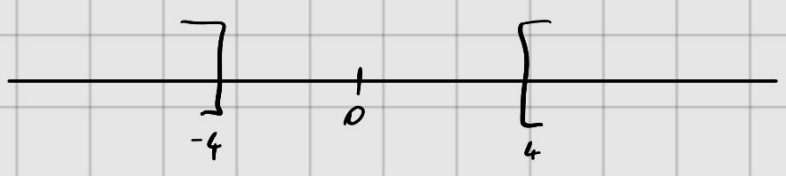
$2(n+1)$

$x-1=w$

$$\sum \frac{(n!)^2}{(2n)!} n^n \qquad \lim_{n\to\infty} \frac{[(n+1)!]^2}{(2n+2)!} \cdot \frac{(2n)!}{(n!)^2} = \lim_{n\to\infty} \frac{(n+1)^2(n!)^2}{(2n+2)(2n+1)(2n)!} \cdot \frac{(2n)!}{(n!)^2}$$

$$(2n+2)! = (2n+2)(2n+1)(2n)! \qquad = \lim_{n\to\infty} \frac{(n+1)^2}{(2n+2)(2n+1)} = \frac{1}{4}$$

$r=4$



$w = |x-1| < 4$ r

ES. 3

$$\sum \frac{(-1)^n x^n}{n+1} \qquad a_n = \frac{(-1)^n}{n+1} \qquad \lim_{n\to\infty} = \left| \frac{a_{n+1}}{a^n} \right| = \ell$$

$n+1$

$$\lim_{n\to\infty}\left|\frac{(-1)}{n+2}\cdot\frac{n+1}{(-1)^n}\right| = \lim_{n\to\infty}\left|\frac{n+1}{n+2}\right| = 1 \qquad (r=1)$$

CONVERGE

$x=1$

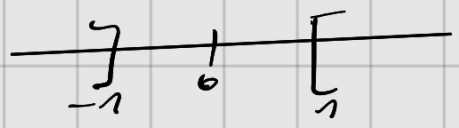
$x=-1$

$\sum \frac{(-1)^n}{n+1}$ CRITERIO DI LEIBNITZ

$|x|<1$

$x=1$

$-1<x<1$

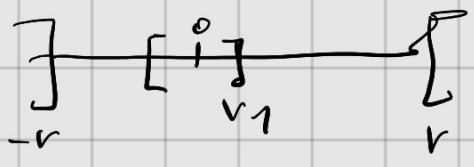


$$\sum \frac{(-1)^n\cdot(-1)^n}{n+1} = \sum \frac{1}{n+1}$$ DIVERGENTE $\qquad]-1,1[$

$$\frac{1}{n^\alpha} = \begin{cases} \text{CON} & \alpha > 1 \\ \text{DIV} & \alpha \le 1 \end{cases}$$

SERIE ARMONICA

TEOREMA: TOTALE CONVERGENZA DELLA SERIE DI POTENZE



$r_1 < r$

$[-r_1, r_1]$ TOT CONV. $\Rightarrow$ U.C.

DATA LA SERIE $\sum a_n x^n$, SIA r IL SUO RAGGIO DI CONVERGENZA, $\forall r_1 < r[$ LE SERIE $\sum a_n r_1^n$ CONVERGE $][-r_1, r_1] \subset]-r, r[$ DOVE LA SERIE È TOTALMENTE CONVERGENTE (UNIFORMEMENTE)

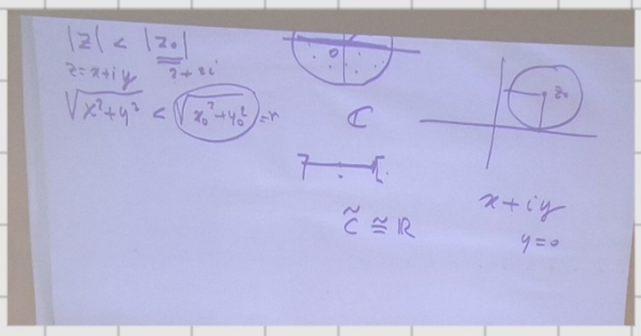
$|a_n x^n| \le$ ESTREMO SUPERIORE $|a_n x^n|$ (TOTALE CONVERGENTE) $\longrightarrow \sum f_n(x)$

se $\sum \sup_I |f_n(x)| < \infty$

$|a_n x^n| \le \sup |a_n x^n| = a_n \sup_I |x^n| \le a_n r_1^n$

$\Rightarrow \sum f_n(x)$ si DEF. TOT CONVERGENTE

$\sum a_n r_1^n \quad r_1 < r$ **UNIFORMEMENTE**



$x^2+1=0$ NON HA SOLUZIONI IN CAMPO $\mathbb{R}$

- TEOREMA DEL LIMITE DI SERIE
- TEOREMA DELL'INTEGRALE DI UNA SERIE
- TEOREMA DELLA DERIVATA DI UNA SERIE

$\sum a_n x^n \quad (\sum f_n(x))$

$$\lim_{x\to\bar{x}} \underbrace{\sum a_n (x-x_0)^n}_{s(x)} = \sum a_n \lim_{x\to\bar{x}} (x-x_0)^n = \sum (\bar{x}-x_0)^n$$

INTEGRALE TEOREMA $(a,b) \subset [-r_1, r_1]$ TOT (U.C.) UNIFORMEMENTE CONVERGENTE

$$\int_a^b \left(\sum a_n (x-x_0)^n\right) dx = \sum a_n \left(\int_a^b (x-x_0)^n dx\right)$$

$$= \sum a_n \left[\frac{(x-x_0)^n}{n+1}\right]_a^b$$

TEOREMA DERIVATA DI UNA SERIE DI POTENZE

$\sum a_n (x-x_0)^n$ CONV. UNIFORMEMENTE CONVERGENTE

$\sum n a_n (x-x_0)^{n-1}$ (U.C.)

$\sum a_n x^n$

TESI $\frac{d}{dx}\left(\sum_{n=0}^{\infty} a_n (x-x_0)^n\right) = \sum_{n=0}^{\infty} a_n \left(\frac{d}{dx}(x-x_0)^n\right)$

$$= \sum a_n \cdot n (x-x_0)^{n-1}$$

$\frac{d}{dx} s(x)$

ES. 1

$$\sum \frac{x^{n+1}}{n+1} = -\log|1-x| \qquad |x|<1 \qquad (\text{U.C.})$$

UNIFORMEMENTE CONVERGENTE

$$\int_0^x \left(\sum_{n=0}^{\infty} t^n\right) dx = \sum \left(\int_0^x t^n \, dt\right) = \sum \frac{x^{n+1}}{n+1}$$

SERIE GEOMETRICA

$$\int_0^x \left(\sum_{n=0}^{\infty} t^n\right) dx = \int_0^x \frac{1}{1-t} \, dt = -\log|1-t| = 0$$

ES. 2

$$\sum \frac{(-1)^n x^n}{n+1} = \sum \frac{(-x)^{n+1}}{n+1} \cdot \frac{1}{(-x)} = \frac{1}{(-x)} \sum \frac{(-x)^{n+1}}{n+1} =$$

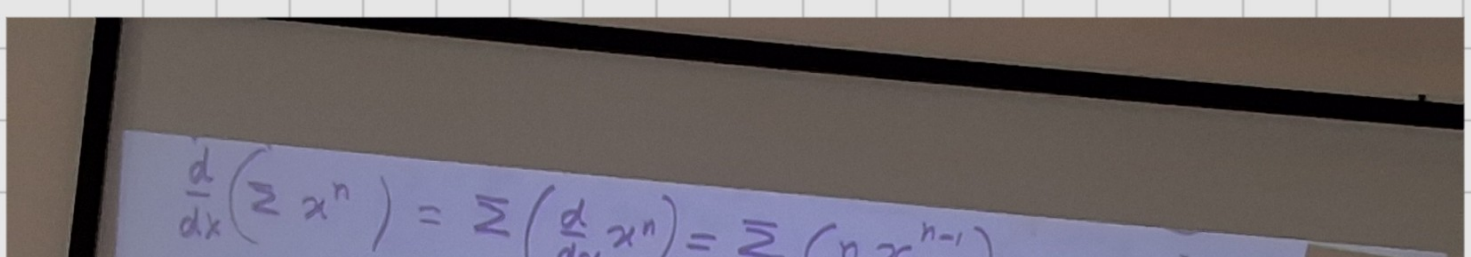
$$= \frac{1}{(-x)} \left[-\log(1+x)\right] = \frac{\log(1+x)}{x}$$

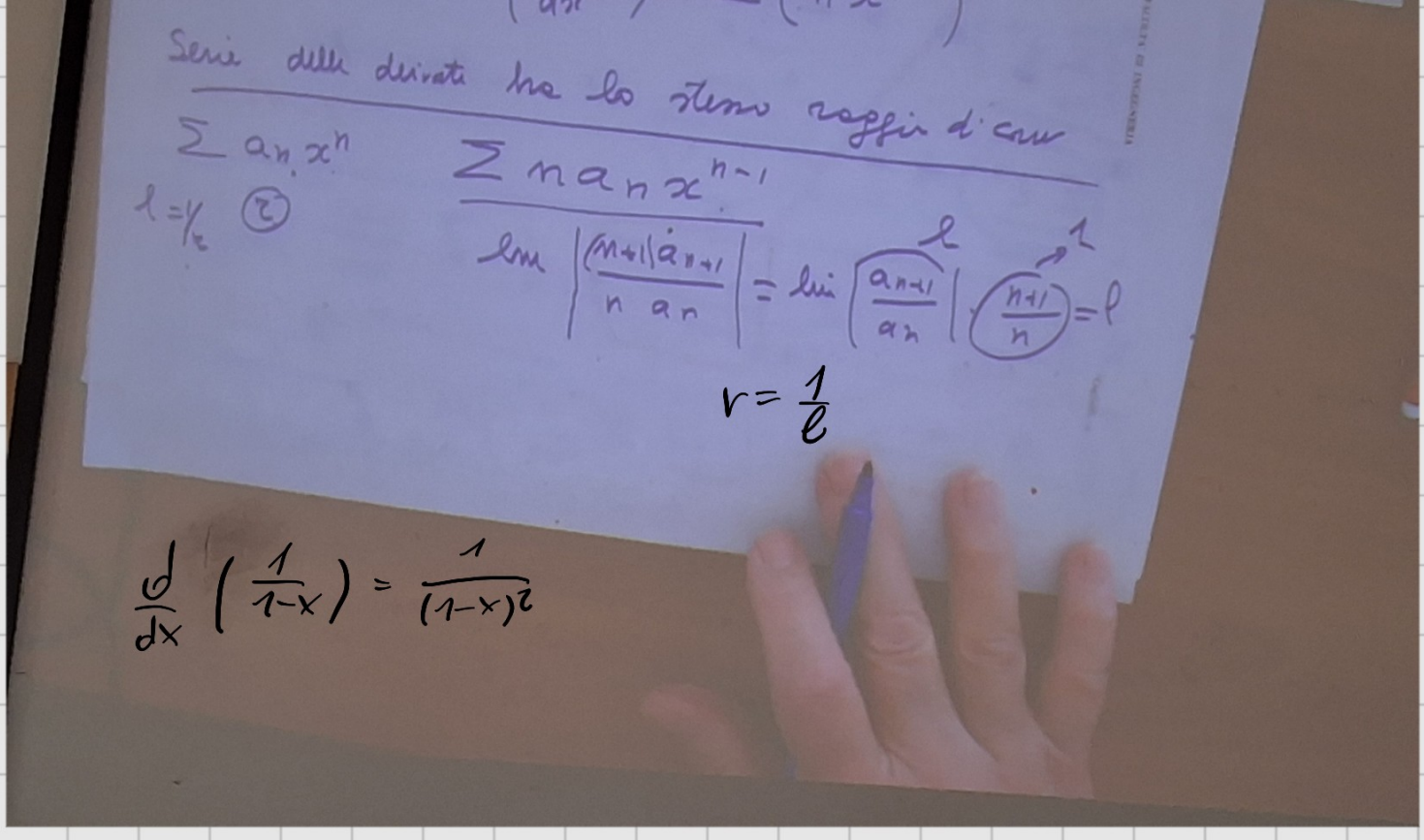
ES. 4

$$\int_0^1 \frac{\log(1+x)}{x} dx = \int_0^1 \left(\sum \frac{(-1)^n x^n}{n+1}\right) dx = \sum \frac{(-1)^n}{n+1} \int_0^1 x^n \qquad \to \frac{x^{n-1}}{n+1}$$

$$S(x) = \sum (-1)^n \frac{1}{(n+1)^2} = \text{FOURIER}$$

ES. 5




Serie delle derivate ha lo stesso raggio di conv
$\sum a_n x^n$
$l = 1/2$ (2)
$\sum n a_n x^{n-1}$
$\lim \left| \frac{(n+1)a_{n+1}}{n\, a_n} \right| = \lim \left| \frac{a_{n+1}}{a_n} \right| \cdot \frac{n+1}{n} = l$
$r = \frac{1}{l}$
$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{1}{(1-x)^2}$